

1.7-10 A CT. SYSTEM

$$x(t) \rightarrow \boxed{S} \rightarrow y(t) = 0.5 \int_{-\infty}^{\infty} x(\tau) [\delta(t-\tau) - \delta(t+\tau)] d\tau$$

Q) WHAT DOES IT DO?

LET'S SOLVE THE INTEGRAL $\rightarrow$ HW? USE THE SIFTING PROPERTY!

USE THE " " PROPERTY } THIS HAPPENS A LOT
USE TABLE " " }

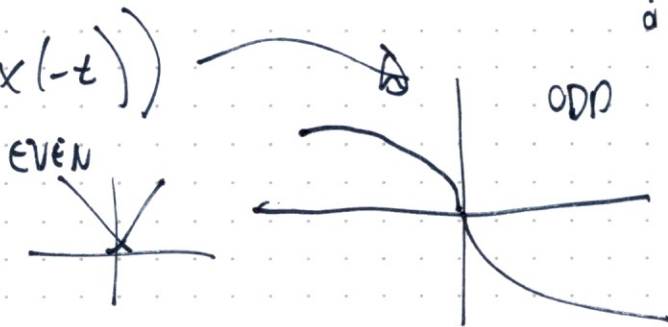
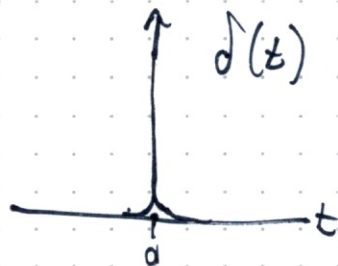
SIFTING PROPERTY $\int f(y) \delta(x-y) dy = f(x)$

IN THE CASE ABOVE

$$y(t) = \frac{1}{2} (x(t) - x(-t))$$

WHAT DOES IT DO?

IT OUTPUTS THE ODD PART OF THE INPUT SIGNAL.



SIFTING PROPERTY BONUS

FOR THE SIGNAL

$$y(t) = 0.5 \int_{-\infty}^{\infty} x(\tau) [\delta(t-\tau) - \delta(t+\tau)] d\tau$$

SPLIT IT INTO TWO PARTS:

$$y(t) = 0.5 \int_{-\infty}^{\infty} x(\tau) \delta(t-\tau) d\tau - 0.5 \int_{-\infty}^{\infty} x(\tau) \delta(t-(-\tau)) d\tau$$

TRICK!

THE SIFTING PROPERTY:

$$\int f(y) \delta(x-y) dy = f(x)$$

APPLYING THAT:

$$y(t) = 0.5 (x(t)) - 0.5 (x(-t))$$

$$y(t) = \frac{1}{2} (x(t) - x(-t))$$

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c) LINEARITY $x(t) \rightarrow \boxed{A} \rightarrow y(t) = \frac{1}{2} (x(t) - x(-t))$

LET'S DEFINE:

$$x_1(t) \rightarrow \boxed{A} \rightarrow y_1(t) = \frac{1}{2} (x_1(t) - x_1(-t))$$

$$x_2(t) \rightarrow \boxed{A} \rightarrow y_2(t) = \frac{1}{2} (x_2(t) - x_2(-t))$$

For $x_3(t) = ax_1(t) + bx_2(t) \rightarrow y_3(t) = \frac{1}{2} ((ax_1(t) + bx_2(t)) - (ax_1(-t) + bx_2(-t)))$

For LINEARITY DOES $y_3(t) = ay_1(t) + by_2(t)$?

$$\begin{aligned} ay_1(t) + by_2(t) &= a \frac{1}{2} (x_1(t) - x_1(-t)) + b \frac{1}{2} (x_2(t) - x_2(-t)) \\ &= \frac{1}{2} [(ax_1(t) + bx_2(t)) - (ax_1(-t) + bx_2(-t))] \end{aligned}$$

$$\therefore ax_1(t) + bx_2(t) \rightarrow ay_1(t) + by_2(t)$$

THE SYSTEM IS LINEAR!

d) MEMORYLESS? DOES THE OUTPUT DEPEND ON PAST OR FUTURE VALUES OF THE INPUT, OR ONLY PRESENT VALUES?

$$y(t) = \frac{1}{2} (x(t) - x(-t))$$

TRY A TEST CASE:

$$t=1 \Rightarrow y(1) = \frac{1}{2} (x(1) - x(-1))$$

It DEPENDS ON A PAST VALUE, so IT IS NOT MEMORYLESS.

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Q) CAUSALITY → DOES THE OUTPUT DEPEND ON FUTURE INPUTS? IF IT DOES THEN THE SYSTEM IS NON-CAUSAL.

CHECK FOR $t > 0$

LET $t=1$ $y(1) = \frac{1}{2}(x(1) - x(-1))$

OUTPUT DEPENDS ON PRESENT + PAST VALUES

CHECK FOR $t < 0$

LET $t=-1$ $y(-1) = \frac{1}{2}(x(-1) - x(1))$

OUTPUT DEPENDS ON PRESENT AND FUTURE VALUES

∴ THE SYSTEM IS NOT CAUSAL

F) TIME INVARIANT? DOES A SHIFT OF THE INPUT RESULT IN THE SAME THING AS SHIFTING THE ORIGINAL OUTPUT?

Text

DEFINE

$$x_1(t) \rightarrow y_1(t) = \frac{1}{2}(x_1(t) - x_1(-t))$$

$$\begin{aligned} x_2(t) = x_1(t+T) &\rightarrow y_2(t) = \frac{1}{2}(x_1(t+T) - x_1(-(t+T))) \\ &= \frac{1}{2}(x_1(t+T) - x_1(-t-T)) \end{aligned}$$

IF IT IS TIME INVARIANT $y_2(t) = y_1(t+T)$

$$y_1(t+T) = \frac{1}{2}(x_1(t+T) - x_1(-t+T))$$

$$\therefore y_2(t) \neq y_1(t+T)$$

THE SYSTEM IS TIME VARIANT!

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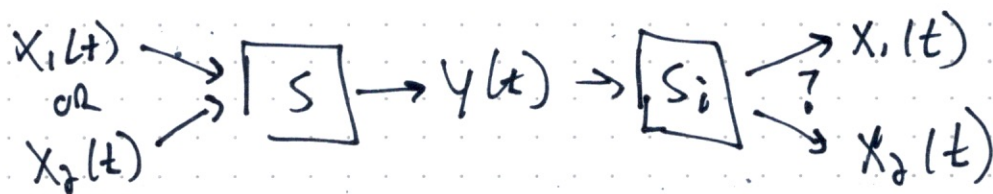
9) INVERTIBLE? $y(t) = \frac{1}{2}(x(t) - x(-t))$

IS THERE SOME OPERATION BY WHICH WE CAN RECOVER THE ORIGINAL INPUT FROM THE OUTPUT?

↑
AND ONLY THE ORIGINAL INPUT!

THE SYSTEM GIVES US THE ODD PORTION OF A SIGNAL...

→ IF TWO DIFFERENT INPUT SIGNALS CAN GIVE US THE SAME OUTPUT, THEN WE CAN'T KNOW WHICH TO RECOVER FROM THE OUTPUT.



CAN TWO DIFFERENT INPUTS GIVE US THE SAME OUTPUT?

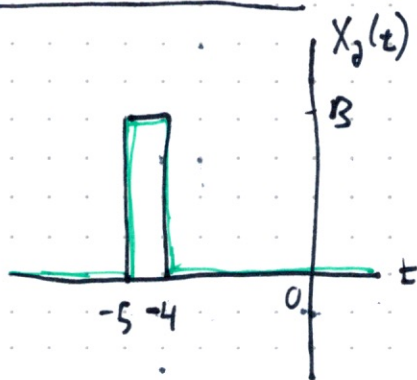
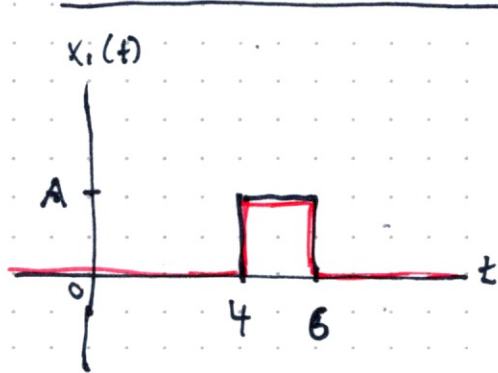
$$x_1(t) = u(t) \rightarrow y_1(t) = \frac{1}{2}(u(t) - u(-t))$$

$$\begin{aligned} x_2(t) = -u(-t) &\rightarrow y_2(t) = \frac{1}{2}(-u(-t) - (-u(-(-t)))) \\ &= \frac{1}{2}(-u(-t) + u(t)) \\ &= \frac{1}{2}(u(t) - u(-t)) \end{aligned}$$

$x_1(t) + x_2(t)$ BOTH PRODUCE THE SAME OUTPUT

∴ THE SYSTEM IS NOT INVERTIBLE.

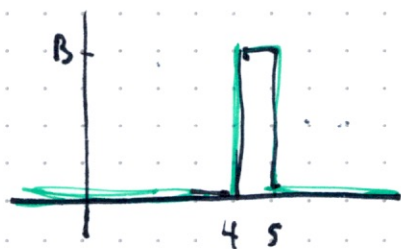
CONVOLUTION - GRAPHICAL APPROACH!



Find:

$$c(t) = x_1(t) * x_2(t)$$

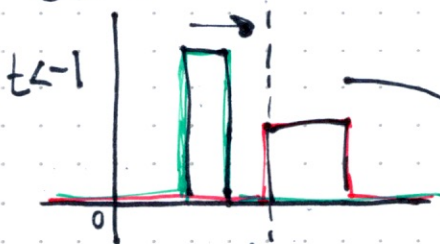
For $x_2(t) \rightarrow x_2(-t)$



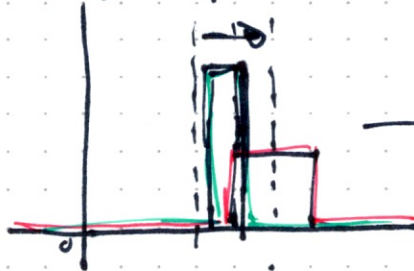
CONSIDER THE 5 DIFFERENT CASES:

- FLIP ONE THEN
- SLIDE + SLIDE
- CALCULATE THE PRODUCT OF THE TWO SIGNALS
- INTEGRATE THE PRODUCT

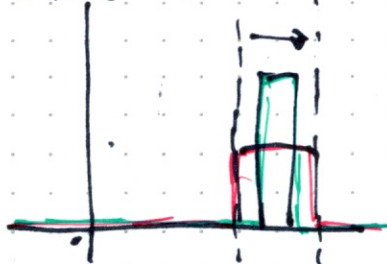
CASE 1:



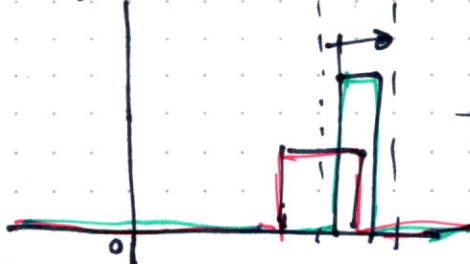
CASE 2: $-1 \leq t < 0$



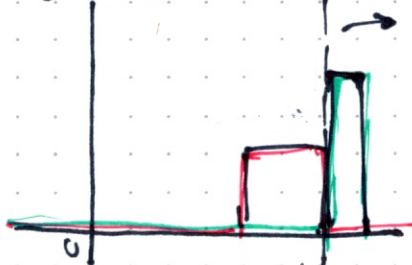
CASE 3: $0 \leq t < 1$



CASE 4: $1 \leq t < 2$



CASE 5: $t \geq 2$



$c(t)$

